

SAT Math

Systems of Linear Equations 2

Question # ID
2.1 cb8f449f

Answer

D

$\frac{1}{2}y = 4$
$x - \frac{1}{2}y = 2$

The system of equations above has solution (x, y) . What is the value of x ?

A. 3

B. $\frac{7}{2}$

C. 4

D. 6

2.2 71189542

C

A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

A. 30

B. 20

C. 19

D. 18

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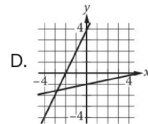
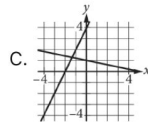
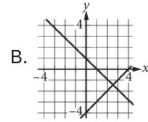
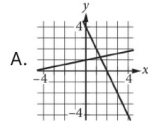
Question # ID
2.3 6e6a3241

$$\begin{aligned}x + 5y &= 5 \\ 2x - y &= -4\end{aligned}$$

Answer

C

Which of the following graphs in the xy -plane could be used to solve the system of equations above?



2.4 fs929f7a

$$\begin{aligned}y &= -\frac{1}{9}x \\ y &= \frac{1}{2}x\end{aligned}$$

C

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -9
- B. -7
- C. 0
- D. 2

2.5 ed92fb68

$$\begin{aligned}4x + 5y &= 100 \\ 5x + 4y &= 62\end{aligned}$$

C

If the system of equations above has solution (x, y) , what is the value of $x + y$?

- A. 0
- B. 9
- C. 18
- D. 38

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Question #	ID		Answer
2.6	19fdf387	In the xy-plane, the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a, b). What is the value of a? A. 3 B. 6 C. 9 D. 12	C
2.7	c5082ce3	The score on a trivia game is obtained by subtracting the number of incorrect answers from twice the number of correct answers. If a player answered 40 questions and obtained a score of 50, how many questions did the player answer correctly?	30
2.8	092ad67d	$\begin{aligned}x + 2y &= 6 \\x - 2y &= 4\end{aligned}$ The solution to the given system of equations is (x, y). What is the value of x? A. 2.5 B. 5 C. 6 D. 10	B
2.9	e77a76ce	Which of the following systems of linear equations has no solution? A. $y = 6x + 3$ $y = 6x + 9$ B. $y = 10$ $y = 10x + 10$ C. $y = 14x + 14$ $y = 10x + 14$ D. $x = 3$ $y = 10$	A

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Question # ID
2.10 5e422ff9

Answer

$$\begin{aligned}y &= 2x - 3 \\ 3y &= 5x\end{aligned}$$

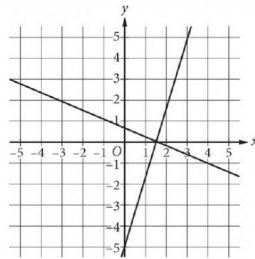
D

In the solution to the system of equations above, what is the value of y ?

- A. -15
- B. -9
- C. 9
- D. 15

2.11 2704399f

A



Which of the following systems of equations has the same solution as the system of equations graphed above?

- A. $\begin{aligned}y &= 0 \\ x &= \frac{3}{2}\end{aligned}$
- B. $\begin{aligned}y &= \frac{3}{2} \\ x &= 0\end{aligned}$
- C. $\begin{aligned}y &= 0 \\ x &= 1\end{aligned}$
- D. $\begin{aligned}y &= 1 \\ x &= 0\end{aligned}$

2.12 b544a348

7

$$\begin{aligned}5x + 3y &= 38 \\ x + 3y &= 10\end{aligned}$$

In the solution (x, y) to the system of equations above, what is the value of x ?

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Question # ID

Answer

2.13 e53688cb

18

$$\begin{aligned}x + 3y &= 29 \\ 3y &= 11\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

2.14 e3bbde69

B

$$\begin{aligned}8x + y &= 5 \\ y &= 9x + 1\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -6
- B. $\frac{4}{17}$
- C. $\frac{6}{17}$
- D. 4

2.15 a0489274

B

$$\begin{aligned}y &= -\frac{1}{5}x \\ y &= \frac{1}{7}x\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of x ?

- A. -5
- B. 0
- C. 2
- D. 7

2.16 7efe5495

C

$$\begin{aligned}y &= 3x \\ 2x + y &= 12\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $5x$?

- A. 24
- B. 15
- C. 12
- D. 5

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Question #	ID		Answer
2.17	0dd6227f	At how many points do the graphs of the equations $y = x + 20$ and $y = 8x$ intersect in the xy-plane? A. 0 B. 1 C. 2 D. 8	B
2.18	2875ba81	_____ $\begin{aligned} 6x + 7y &= 28 \\ 2x + 2y &= 10 \end{aligned}$ The solution to the given system of equations is (x, y). What is the value of y? A. -2 B. 7 C. 14 D. 18	A
2.19	686b7cad	A proposal for a new library was included on an election ballot. A radio show stated that 3 times as many people voted in favor of the proposal as people who voted against it. A social media post reported that 15,000 more people voted in favor of the proposal than voted against it. Based on these data, how many people voted against the proposal? A. 7,500 B. 15,000 C. 22,500 D. 45,000	A
2.20	65833256	$\begin{aligned} y &= 6x + 16 \\ -7x - y &= 36 \end{aligned}$ What is the solution (x, y) to the given system of equations? A. $(-4, -8)$ B. $(-\frac{20}{13}, -\frac{80}{13})$ C. $(4, 40)$ D. $(20, 136)$	A

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Question # ID
2.21 dcc4886a

Answer

D

$$y = \frac{2}{7}x + 3$$

One of the two equations in a system of linear equations is given. The system has infinitely many solutions. If the second equation in the system is $y = mx + b$, where m and b are constants, what is the value of b ?

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

2.22 6e50ce28

A

The sum of a number x and 7 is twice as large as a number y . The number y is 3 less than the number x . Which system of equations describes this situation?

- A. $x + 7 = 2y$
 $y = x - 3$
- B. $x + 7 = 2y$
 $y = 3 - x$
- C. $2(x + 7) = y$
 $y = x - 3$
- D. $2(x + 7) = y$
 $y = 3 - x$

2.23 bd45df49

D

$$\begin{aligned}y &= 3x + 9 \\ 3y &= 8x - 6\end{aligned}$$

The solution to the given system of equations is (x, y) . What is the value of $x - y$?

- A. -123
- B. -33
- C. 3
- D. 57